LLM-powered smart contract auditor.

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Repository: github.com/ulan-mustafin/csci435-smart-contract-auditor

Objective:

Build an automated evaluation pipeline using LLMs to scan solidity contracts for known vulnerabilities, including reentrancy and integer overflow/underflow, and benchmark performance against a labeled dataset.

Scope:

In-Scope

- Detecting reentrancy and integer overflow/underflow in Solidity source files.
- Comparing prompt variants.
- Testing on at least 50 smart contracts and calculating precision on each vulnerability type.

Out-Of-Scope:

- Training the model; only prompting the existing LLMs.
- Gas optimization.
- Code quality, documentation and business logic checks.
- Bytecode level analysis.
- Automatic patch and exploit generation.

Key Risks:

Label noise, unstable output, LLM hallucination triggered false alarms and training data leakage.

Dataset and Evaluation:

At least 50 labelled Solidity contracts, including at least 15 safe ones. We score precision and recall for each vulnerability type and compare against Slither as the baseline. A results table will show which prompt gives the best accuracy.

Deliverables:

The Github repository with the pipeline an dataset, the results table, an error analysis, and the final report.